



Combustion chemistry of ammonia/hydrogen mixtures: Jet-stirred reactor measurements and comprehensive kinetic modeling

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ABSTRACT

To investigate the oxidation of ammonia (NH₃)/hydrogen (H₂) mixtures at intermediate temperatures, this work has implemented jet-stirred reactor (JSR) oxidation experiments of NH₃/H₂ mixtures at atmospheric pressure and over 800–1280 K. The H₂ content in the NH₃/H₂ mixtures is varied from zero to 70 vol% at equivalence ratios of 0.25 and 1.0. Species identification and quantification are achieved by using Fourier-transform infrared (FTIR) spectroscopy. A kinetic model for pure NH₃ and NH₃/H₂ mixtures is also developed for this research, and validated against the present experimental data for pure NH₃ and NH₃/H₂ mixtures, as well as those for pure NH₃, H₂/NO, H₂/N₂O, NH₃/NO, NH₃/NO₂ and NH₃/H₂ mixtures in literature. The model basically captures the experimental data obtained here, as well as in literature. Both measured and predicted results from this work show that H₂ blending enhances the oxidation reactivity of NH₃. Based on the model analysis, under the present experimental conditions, NH₃ + H = NH₂ + H₂ proceeds in its reverse direction with increasing H₂ content. The H atom produced is able to combine with O₂ to produce either O and OH via a chain-branching reaction, or to yield HO₂ through a chain-propagation reaction. HO₂ is an important radical under the present intermediate-temperature conditions, which can convert NH₂ to OH via NH₂ + HO₂ = H₂NO + OH; H₂NO is then able to convert H to NH₂ and OH. In this reaction sequence, NH₂ and H₂NO are chain carriers, converting HO₂ and H to two OH radicals. Since the OH radical is the dominant radical to consume NH₃ under the present conditions, the enhanced OH yield via H + O₂ = O + OH, NH₂ + HO₂ = H₂NO + OH and H₂NO + H = NH₂ + OH, with increasing H₂ content, promotes the consumption of NH₃. For NO_x formation, non-monotonous trends are observed by increasing the content of H₂ at the 99% conversion of NH₃. These trends are determined by the competition between the dilution effects and the chemical effects of H₂ addition. Nitrogen related radicals, such as NH₂, NH and N, decrease as H₂ increases, and this dilution effect reduces NO_x formation. For chemical effects, the yields of oxygenated radicals, such as O, OH and HO₂, are enhanced with increasing H₂ content, which results in enhancing effects on NO formation. For N₂O formation, the enhanced oxygenated radicals (O, OH and HO₂) suppress its formation, while the enhanced NO promotes its formation.

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1. Introduction

To mitigate global warming and achieve carbon neutrality by mid-century, variable renewable energy sources (VRES) must scale up to increase their shares in the total primary energy supply in the near future [1]. Among the VRESs, ammonia (NH₃) is an attractive carbon-free fuel with zero carbon emissions in its combustion process; it is an effective hydrogen (H₂) carrier that can store 17.8% of H₂ by weight and which exhibits a volumetric en-

ergy density of 11.3 GJ/m³ in liquid form at 1.1 MPa and 300 K [2]. The past decades have firmly established the production, transport and storage infrastructures of NH₃ [3,4], strengthening its potential applications as a fuel candidate or an H₂ carrier. However, as a fuel candidate, NH₃ exhibits low laminar flame speed, high auto-ignition temperature and a narrow flammability range, which are main challenges to its practical applications [4].

H₂ (which can be considered as another VRES [5,6]), also produces zero carbon emissions, while its high reactivity associated with a wide flammability range [7] and high flame speed [8] causes considerable concern about its practical storage and utilization. Considering the respective properties of NH₃ and H₂,

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cofiring NH_3/H_2 mixtures produces zero carbon emissions, while enhancing the combustion intensity of NH_3 and diminishing the safety concerns regarding H_2 . For this reason, more attention is directed towards NH_3/H_2 cofiring, with increasing research focusing on engine performance when fueled with NH_3/H_2 mixtures [9–12].

Fundamental combustion chemistry studies are helpful to better understand the effects of NH_3/H_2 blending and the inherent mechanism behind engine performance. Recent fundamental experimental studies on NH_3/H_2 mixtures have focused mainly on global combustion parameters—laminar flame speeds [13–20] and ignition delay times [21–24]. Among these studies, Wang et al. [13] reviewed the measured laminar flame speeds of NH_3/H_2 mixtures before 2020 and contributed to a new set of experimental data using a heat flux method at an unburned temperature (T_u) of 298 K. Their measured laminar flame speeds covered an unburned pressure (P_u) range of 1–5 atm and equivalence ratios (ϕ) of 0.7–1.6. Lhuillier et al. [15] measured the laminar flame speeds of NH_3/H_2 mixtures at atmospheric pressure using the outwardly propagating spherical flame method. The T_u covered 298–473 K, and H_2 content (referred to as $x(\text{H}_2)$ in this work), ranged from 0–60 vol%. Ichikawa et al. [17] used the same method to obtain the flame speeds of stoichiometric NH_3/H_2 mixtures at room temperature and $P_u = 1$ –5 atm. The H_2 content covered 0–100 vol%. Recently, Shrestha et al. [14] measured the laminar flames of NH_3/H_2 mixtures at evaluated temperatures of 298–473 K and pressures of 1–10 atm in a constant volume combustion chamber, with the H_2 content varying from zero to 30 vol%. In summary, laminar flame speeds of NH_3/H_2 mixtures are available under a wide range of temperatures, pressures, equivalence ratios and H_2 content conditions in the literature, providing various high-temperature oxidation data for model validation. In the literature, the experimental data for low- and intermediate-temperature oxidation were measured in rapid compression machines (RCMs) and in shock tubes. The ignition delay times in RCMs at high pressures were reported by Dai et al. [22], Pochet et al. [23] and He et al. [24], covering a pressure range of 20–75 atm, temperatures of 950–1210 K, equivalence ratios of 0.2–2.0 and H_2 content of zero to 25 vol%. Chen et al. [21] recently measured the ignition delay times of stoichiometric NH_3/H_2 mixtures using a shock tube at temperatures from 1020 to 1945 K, pressures of 1.2 and 10 atm, and hydrogen fractions from zero to 70%. Compared with global parameters, the speciation data of NH_3/H_2 mixtures can provide more stringent constraints for model development, especially for NO_x formation. To the best of our knowledge, only the speciation data of NH_3/H_2 mixtures measured under flame conditions are available in the literature [25–27], therefore, additional speciation data measured at intermediate temperatures, are extremely valuable for model validation.

A pure NH_3 model includes the sub-mechanism of H_2 , and thus can be used to predict the combustion behavior of NH_3/H_2 mixtures. A number of NH_3 kinetic models are available in the literature of the past five years; most of them focused on particular combustion conditions and were validated against limited experimental data. Kawka et al. [28] recently compared the performance of eight NH_3 models [29–36] against the homogeneous NH_3 combustion data; they found that none of the selected models could reasonably predict all the selected experimental data. Therefore, comprehensive kinetic modeling studies on pure NH_3 and NH_3/H_2 mixtures are highly desirable.

In this work, jet-stirred reactor (JSR) oxidation experiments on NH_3/H_2 are conducted at atmospheric pressure, with H_2 content ranging from zero to 70 vol%. The temperature ranges cover 800–1280 K, and equivalence ratios are at 0.25 and 1.0. Speciation data of NH_3 , H_2O , NO and N_2O are measured using Fourier-transform infrared (FTIR) spectroscopy. A comprehensive kinetic model for NH_3 and NH_3/H_2 is developed, based mainly on the evaluation of

kinetic parameters, and validated against the present JSR experimental data and those available in the literature. Based on this model, the blending effects of H_2 on NH_3 oxidation under present conditions are revealed, including both fuel conversion and NO_x formation.

2. Experimental methods

The JSR experiments are conducted at King Abdullah University of Science and Technology (KAUST). Figure 1 shows the schematic of the experimental setup. The JSR used in this study is made from quartz, with a spherical reactor volume of 76 cm^3 , and two 18 cm long stems on both sides for the inlet and outlet. To achieve homogenous mixing, four opposing nozzles, with an internal diameter of 0.3 mm, are used to create a jet flow stirring. The JSR is kept inside an electric tube furnace (CY-O1200-50ICS) with a safe working temperature of ≤ 1425 K. Temperatures inside the reactor are measured with a K-type thermocouple inserted into the reactor. Experiments include the oxidation of pure NH_3 and NH_3/H_2 mixtures at atmospheric pressure; reactants are introduced into the spherical reactor through a quartz inlet tube, using nitrogen as a carrier gas. To avoid pre-reactor oxidation, the oxygen diluted with nitrogen is added to the reactor through an outer channel and mixed only within the reactor sphere. All the gases except nitrogen are from cylinders with 99.999% purity. Nitrogen is supplied from a central unit of the building with 99.99% purity. The total initial mole fraction of NH_3/H_2 mixtures is 2000 ppm, with H_2 content varying from zero to 70 vol%. The amount of oxygen required in the experiments varies depending on equivalence ratios of 0.25 and 1.0. The equivalence ratios are calculated with $\phi = (\frac{\text{NH}_3}{\text{O}_2})/(\frac{\text{NH}_3}{\text{O}_2})_{\phi=1}$ based on the global reaction of $\text{NH}_3 + 1.25\text{O}_2 = \text{NO} + 1.5\text{H}_2\text{O}$. This definition for equivalence ratio calculation was also adopted in recent JSR experiments from Dagaut [37]. The residence time (τ) is set to 1 s. Table 1 summarizes the experimental conditions. The flow rates of NH_3 , H_2 , O_2 and N_2 are controlled with MKS mass flow controllers and a multi-gas controller, and are adjusted with the increasing reaction temperatures to maintain constant residence time.

The products are extracted through a quartz probe from the JSR and sent directly to the gas cell of FTIR (Thermo-Scientific, Nicolet 6700) via a heated line maintained at 403 K. The gas cell has a 200 ml volume and 2 m optical pathlength; it is heated to 403 K to avoid condensation of gaseous species within the cell. An external pump is attached to the outlet of the FTIR gas cell; it operates continuously to remove gases from the JSR through the cell. The FTIR is purged with nitrogen throughout the experiment. Three to four spectra are collected for each temperature. For the experiments, the mole fractions of NH_3 , H_2O , NO and N_2O are obtained based on the calibration experiments for each species; their experimental uncertainties are estimated to be within 10%. The mole fraction profiles of NH_3 , H_2O , NO and N_2O , under various experimental conditions, are provided in the *Supplementary Material*.

3. Kinetic modeling

The present ammonia model is constructed based on a recent study by Mei et al. [38]. In the Mei model, the sub-mechanism of H_2 from Glarborg's research group [39] is adopted, which has been validated against comprehensive experimental targets of hydrogen combustion. In addition, several chemically termolecular reactions (represented as $\text{H} + \text{O}_2 + \text{H}/\text{O}/\text{OH} = \text{products}$), are added into the base hydrogen model, as they were not included in the H_2 mechanism from Glarborg's research group [39]. Rate constants for the chemically termolecular reactions are adopted from a theoretical calculation study from Burke and Klippenstein [40].

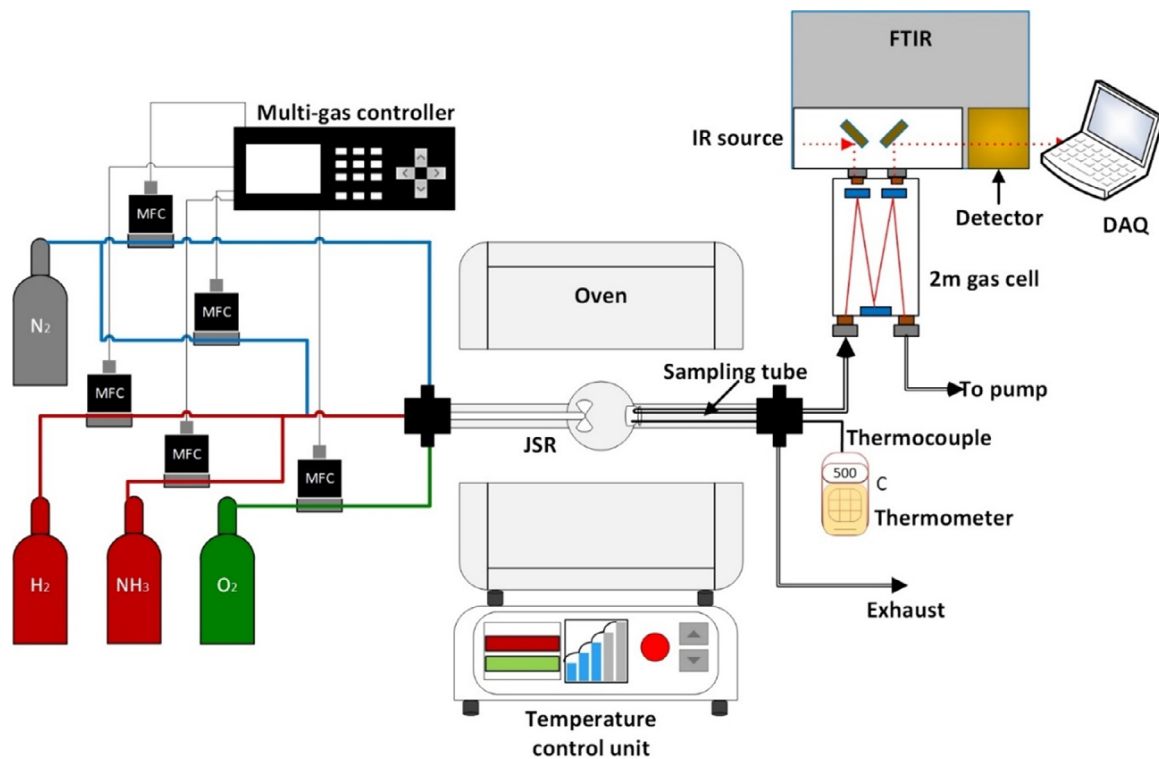


Fig. 1. Schematic of JSR experimental setup for the oxidation of NH_3 and NH_3/H_2 mixtures.

Table 1
Experimental conditions in the present work.

$x(\text{H}_2)\%$	$\text{NH}_3(\text{ppm})$	$\text{H}_2(\text{ppm})$	$\text{O}_2(\%)$	$\text{N}_2(\%)$	τ/s	$T(\text{K})$	P/atm	ϕ
0	2000	0	1	98.8	1	900–1280	1	0.25
0	2000	0	0.25	99.55	1	900–1280	1	1.0
10	1800	200	0.94	98.86	1	900–1280	1	0.25
10	1800	200	0.235	99.565	1	900–1280	1	1.0
30	1400	600	0.82	98.98	1	900–1220	1	0.25
30	1400	600	0.205	99.595	1	900–1220	1	1.0
50	1000	1000	0.7	99.1	1	800–1170	1	0.25
50	1000	1000	0.175	99.625	1	800–1120	1	1.0
70	600	1400	0.58	99.22	1	800–1060	1	0.25
70	600	1400	0.145	99.655	1	800–1060	1	1.0

According to the previous study [41], the incorporation of these termolecular reactions lowers the predictions of laminar flame speeds of hydrogen and captures the experimental data better. The base model is then validated against the experimental data of H_2/NO [42,43] and $\text{H}_2/\text{N}_2\text{O}$ [44,45] mixtures to verify the reliability of the sub-mechanisms of NO and N_2O (two important intermediates in the ammonia oxidation). Validation results can be found in Figs. S1–S8 in the *Supplementary Material*. Generally, the base model predicts these experimental targets well. The sub-mechanism of NH_3 is revisited in this work. Thermodynamic data of NH , NH_2 , NNH and N_2H_2 are updated based on recent theoretical calculations from this group [46], and the rate constants of important reactions are evaluated based on experimental [47,48], theoretical [46,49–53] and review [3,31,54] studies.

3.1. Important reactions in the (oxidative) pyrolysis of NH_3

The (oxidative) pyrolysis of ammonia can be divided into N1 and N2 pathways. The N1 pathways are through $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NH} \rightarrow \text{N}$. Critical reactions along these pathways were recently evaluated by Elishav et al. [3], which includes

unimolecular decomposition reactions and the H-atom abstraction reactions. R1 and R2 are two unimolecular decomposition reactions of ammonia. Since the branching ratio of R2 is only 0.025 (according to Elishav et al. [3]), only R1 is incorporated in this model. The rate constant of R1 is adopted from the measured results of Davidson et al. [47] in a shock tube. For the H-atom abstraction reactions by H (R3), the rate constant computed by Nguyen et al. [52] agrees well with the measurements in the literature, and it is 30% lower than that calculated by Stagni et al. [49] at above 1000 K. Their computed rate constant was recommended by Elishav et al. [3] and adopted in this model. Rate constants of other important H-atom abstraction reactions by OH, HO_2 and O (R4–R6) are adopted from recent calculations from Stagni et al. [49]. The rate constant of R7 is taken from recent calculation work by this group, with a high level of theory [53], which agrees within a factor of 2 with recent theoretical calculation [49] and measured results [48]. The conversion of NH_2 to NH occurs mainly through the H-atom abstraction reactions. Among them, the rate constants of H-atom abstraction reactions of NH_2 by H and OH radicals (R8 and R9), are adopted from results computed by this group [46]. Similarly, the conversion of NH to N also occurs mainly via H-atom abstraction reactions such as R10

and R11; their rate constants are adopted from Yao et al. [55] and Klippenstein et al. [56], respectively.



Like methane combustion (particularly under fuel-rich conditions), NH_2 and NH are likely to proceed along N_2 pathways to produce N_2 species. According to the theoretical calculations of Klippenstein et al. [56], the potential energy surface (PES) of $\text{NH}_2 + \text{NH}_2$ has different reaction channels. At high pressure and low temperature conditions, R12 becomes competitive. The rate constant of R12 is taken from their theoretical calculation work, while the third-body collision efficiencies of N_2 , Ar, O_2 and NH_3 are taken from a recent experimental study of Glarborg et al. [48]. In addition to $\text{NH}_2 + \text{NH}_2$ potential, the PES of $\text{NH}_2 + \text{NH}$ was also theoretically investigated by Klippenstein et al. [56]. The rate constants of various competitive reaction channels on this PES are initially adopted from the calculations of Klippenstein et al. [56]. To better capture the laminar flame speeds of rich ammonia/air mixtures, and the speciation data of NH_2 and NH in ammonia pyrolysis, the original rate constant of R13 is reduced by a factor of three. The rate constants of H-atom abstraction reactions of N_2H_2 by H and O atoms are adopted from a recent theoretical calculation study by this group [46], while those from the OH radical is adopted from theoretical work of Raghunath et al. [57]. The subsequent dominant reaction of NNH is the unimolecular decomposition reaction to produce N_2 and H (i.e. R14); its rate constant still has large discrepancies [32]. In this work, the rate constant of R14 is referred to the measured value from [58]. Since these reactions are critical to predicting the speciation data measured in the pyrolysis of NH_3 and hydrazine (N_2H_4), shock tube pyrolysis experiments of NH_3 from Davidson et al. [47] and flow reactor pyrolysis experiments of N_2H_4 from Halat-Augier [59] are selected as validation targets to verify the (oxidative) pyrolysis mechanism of NH_3 . The validation results can be found in Figs. S9 and S10 in the *Supplementary Material*. In general, the present model predicts these experimental data well.



3.2. Important reactions involved in DeNO_x pathways

The most important DeNO_x pathways in ammonia oxidation are R15 and R16, converting NO to NNH and N_2 , respectively. These two reactions are also responsible for chain branching and chain termination in ammonia oxidation, thus, they present great sensitivity under most combustion conditions. Klippenstein et al. [32] evaluated the overall rate constant and the branching ratio of $\text{NH}_2 + \text{NO}$. The uncertainties of the overall rate constant are generally within a factor of two; in this work, their evaluated rate constants are adopted in the present model. Similarly, NH_2 can combine NO_2 to proceed both chain-branching (i.e. R17) and chain-termination reactions (i.e. R18). The overall rate constant and the branching ratio of $\text{NH}_2 + \text{NO}_2$ were evaluated by Glarborg et al. [31]. Their recommended overall rate constant [31] is adopted in this model, which lies between the computed value from Klippenstein et al. [60] and the measured value from Song et al. [61]. The branching ratio of R18 (defined as $k_{18}/(k_{18}+k_{17})$), is about 20%, according to the evaluation of Glarborg et al. [31], which is also adopted in this model. In addition to NH_2 , NH can convert NO to N_2O and N_2 via R19 and R20, respectively, both of which are responsible for chain propagation. Their rate constants are adopted from the evaluation work of Baulch et al. [62] in this model, since their evaluated results agree well with the measured results in the literature. The reactions mentioned above play an important role in the DeNO_x process, and therefore the experimental data of NH_3/NO [37,63–68] and NH_3/NO_2 mixtures [69] are selected as the validation targets to verify the DeNO_x related mechanism. The model validation results against the speciation data in JSRs and flow reactors, as well as the laminar flame speed data, are presented in Figs. S11–S19 in the *Supplementary Material*. Generally, the present model reasonably predicts all the experimental data from the literature.



3.3. Important reactions involved in NO_x formation pathways

The NO_x formation reaction pathways occur mainly through the oxidation reactions of NH_2 , NH and N . These radicals may react with O_2 , O, OH and HO_2 to produce NO, or oxygenated intermediates, such as H_2NO and HNO , which can eventually be converted to NO. At high temperatures, NH_2 can combine with the O atom to produce HNO and H via R21, which is more favorable than its competitive reaction channel, R22 [31]. Experimental studies on R21 and R22 are only available at low temperatures, thus the temperature dependencies of these two reactions still include considerable uncertainty. In this work, the computed rate constant for R21 from Sumathi et al. [70], and the measured rate constant for R22 from Inomata et al. [71] are adopted. Future studies are still necessary to obtain their rate constants at high temperatures. At intermediate and low temperatures, the HO_2 radical becomes more important. The combination between NH_2 and HO_2 can generate H_2NO and OH via R23. In this way, the HO_2 radical can be converted to the more reactive OH radical. The rate constant of R23 is still very uncertain at room temperatures [48]. The evaluation value from Song

et al. [72] is adopted in this work; it is close to the upper limit measured by Cheskis and Sarkisov [73] and agrees within a factor of 2 with recent experimental study by Glarborg et al. [48]. Additional studies are still necessary to reduce the uncertainties of this reaction. NH may also be oxidized by O₂, O and OH to produce NO and HNO. The rate constants of these reactions were recently reviewed by Elishav et al. [3]. The rate constant of R24 is referred to the measured result from Römmeing et al. [74], while that of the R25 is adopted from the evaluated value by Elishav et al. [3] in this work. The rate constants of R26 and its competitive H-atom abstraction reaction are taken from the computed results of Klippenstein et al. [56], while the rate constants of R27 and its competitive H-atom abstraction reaction are from Guadagnini et al. [75].



The sub-mechanism of H₂NO still contains great uncertainties. The rate constant of R28 is adopted from the evaluation value of Glarborg et al. [31], in which the activation energy of R28 was lowered by 2.5 kcal/mol, based on a theoretical calculation study of Song et al. [72]. Other bimolecular reaction pathways and their rate constants involving H₂NO are adopted from earlier calculation work of Dean and Bozzelli [76]. One of the most important consumption reactions of HNO is the unimolecular decomposition reaction, leading to the formation of NO and H (i.e. R29). This is an important source for NO formation. The rate constant of this reaction is adopted from recent theoretical calculations of Stagni et al. [49]. HNO may also be consumed by reactive radicals via bimolecular reactions, such as R30-R32. The rate constants of R30-R32 are adopted from Soto et al. [77], Miller et al. [54] and Inomata et al. [71], respectively. These reactions play an important role in pure NH₃ oxidation, therefore, the experimental data of pure NH₃ oxidation are used to validate the sub-mechanism of NH₃ oxidation, including the speciation data measured in JSRs [37,49] and flow reactors [49,72] and the global parameters like ignition delay times [78,79] and laminar flame speeds [14,15,38,80]. The validation results can be found in Figs. S20-S34 in the *Supplementary Material*. In general, the present model satisfactorily captures all these experimental data.



3.4. Simulation method

All the simulations in this work are performed using ANSYS CHEMKIN-PRO software [81]. Jet-stirred reactor (JSR) experiments

are simulated using a transient perfectly stirred reactor. To achieve a steady-state solution, the simulation end times are set at 20 times greater than the mean residence times. Speciation profiles measured in shock tubes are simulated with a closed homogeneous batch reactor. Adiabatic and constant volume conditions are assumed, and the reflected temperature (T₅) and pressure (P₅) are set as initial temperature and pressure. For flow reactor simulations, both the closed homogeneous batch reactor and the plug flow reactor are used, depending on the boundary conditions. Speciation data measured in laminar premixed flat flames are simulated by the premixed laminar burner-stabilized flame module. Mass flow rates and temperature profiles provided in the experiments are used as boundary conditions. Shock tube and RCM ignition delay time simulations are performed using a closed homogeneous batch reactor with adiabatic and constant-volume assumptions. The volume profiles in particular, are incorporated in RCM simulations to account for the heat loss effects. Laminar flame speeds are simulated using the premixed laminar speed calculation module. Thermal diffusion and mixture-averaged transport equations are accounted for in the simulations. To achieve grid-independent results, GRAD and CURV values are assigned to be 0.1 or smaller during the simulations. All the validation data are summarized in Table S1 in the *Supplementary Material*. The kinetic, thermodynamic and transport files of the present model are provided in the *Supplementary Material*.

4. Results and discussion

4.1. Measured and predicted results

The present model is first validated against the JSR oxidation data of pure H₂ from Dayma and Dagaut [43] at 1 atm and equivalence ratios of 0.2 and 1.0. The comparison between their measured and present predicted results is shown in Fig. 2. The present model captures the decomposition of H₂ and the formation of H₂O well. The performance of the present model is also compared against the JSR oxidation experiments of pure NH₃ from Stagni et al. [49] and Dagaut [37] in Figs. S20-S24 in the *Supplementary Materials*; it shows that the present model satisfactorily captures the consumption of NH₃ and the formation of products.

The measured results of the present JSR oxidation experiments of NH₃/H₂ mixtures at equivalence ratios of 0.25 and 1.0 are presented in Figs. 3 and 4. To better compare the H₂ blending effects, the measured mole fractions of NH₃ are converted to the conversion ratios (ω) using Eq. 1, where x(T₀) and x(T_i) represent the initial mole fraction of NH₃ at the initial temperature, and the NH₃ mole fraction measured at each temperature, respectively. As seen from Figs. 3a and 4a, the conversion ratio of NH₃ is increased at given temperatures and the temperature window of NH₃ consumption is lowered with the increasing content of H₂. Both phenomena reveal that H₂ blending enhances the oxidation of NH₃. To interpret the experimental data, the present model, and four recent models from different research groups are selected to predict the present experimental data. These literature models are named the NUI 2017 model [30], the Glarborg 2018 model [31], the Polimi 2020 [49] and the Nakamura 2017 model [82], respectively. The simulated mole fractions of NH₃ are also converted to the conversion ratios for comparison. The predicted results by the present, Polimi 2020 [49] and Glarborg 2018 models [31] are presented in Figs. 3 and 4, while those by the NUI 2017 [30] and Nakamura 2017 model [82] are shown in Figs. S35 and S36 in the *Supplementary Materials*. As seen from Figs. 3a, 4a, S35a and S36a in the *Supplementary Materials*, all the models, except the Nakamura 2017 model [82] are able to capture the enhanced effects of H₂ blending. At fuel-lean conditions, the predicted conversion of NH₃ and H₂O in pure NH₃ is faster than that in the 90%NH₃/10%

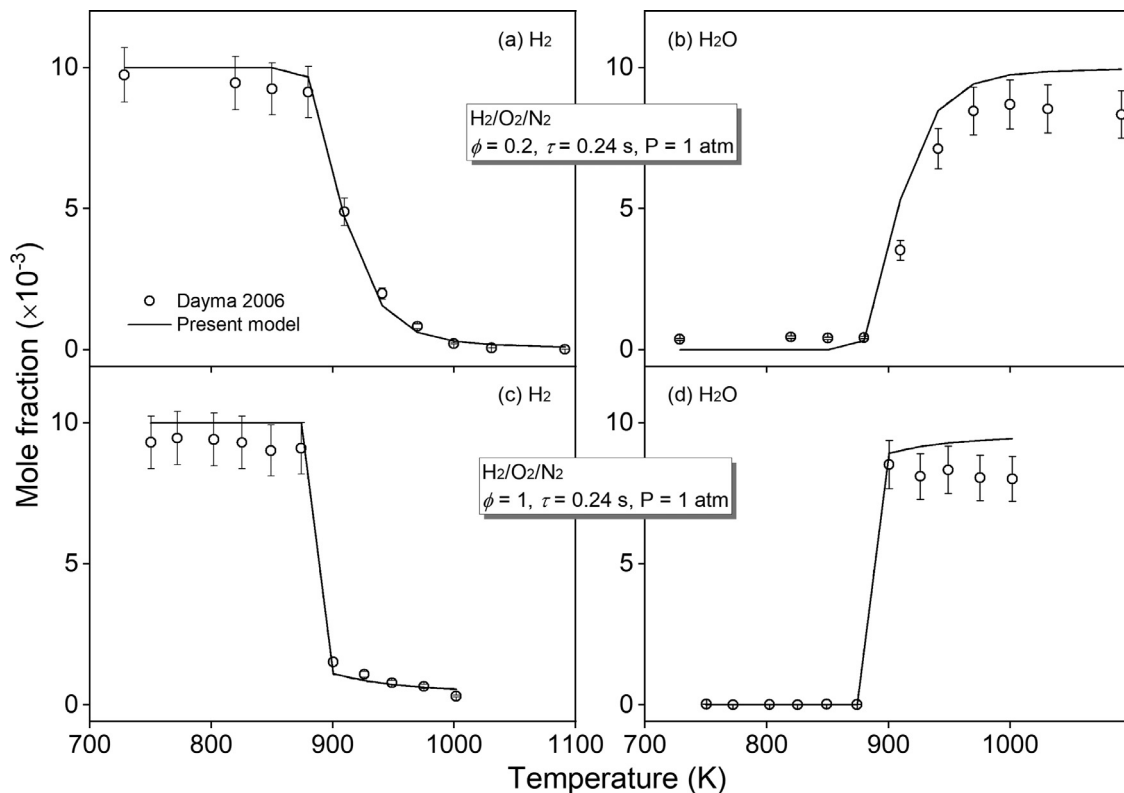


Fig. 2. Measured and simulated mole fractions of (a) H_2 and (b) H_2O in the JSR oxidation of pure H_2 at $\phi = 0.2$, $\tau = 0.24$ s and $P = 1$ atm; (c) H_2 and (d) H_2O in the JSR oxidation of pure H_2 at $\phi = 1.0$, $\tau = 0.24$ s and $P = 1$ atm. Symbols are measured results from Dayma and Dagaut [43]. Lines are simulated results from the present model.

H_2 mixture using the Nakamura 2017 model [82], which is contradictory to the present experiments, and to the performance of the other models. Among the models, the present model can best capture the temperature windows of NH_3 conversion measured under fuel-lean conditions; Polimi 2020 model [49], NUI 2017 model [30] and the present model can reasonably predict the oxidation rates of NH_3 with increasing H_2 content under stoichiometric conditions.

$$\omega = \frac{x(T_0) - x(T_i)}{x(T_0)} \times 100\% \quad (1)$$

The measured and simulated NO and N_2O formation are presented in Figs. 3(c,d), 4(c,d), S35(c,d) and S36(c,d) in the *Supplementary Materials*. Their measured onset temperatures are lowered by increasing the H_2 content, which is consistent with the temperature windows of NH_3 conversion. Also, under the same conditions, the measured NO and N_2O mole fractions are comparable. All the models predict the yields of NO reasonably under varying conditions, while over-predicting the formation of N_2O except the present model. It is noted that the Polimi 2020 model largely over-predicts the yields of N_2O measured in the present experiments, as well as those measured in $NH_3/NO/NO_2$ mixtures and in pure NH_3 experiments (see Figs. S14, S17, S18 and S22 in the *Supplementary Materials*). In contrast, the present model captures the formation of N_2O measured in the present experiments and those measured in the literature [37,66,69]. In addition, the signals of NO_2 are not observed in the present experiments, indicating its low concentration level under the present operation conditions. The NO_2 concentrations predicted by various models are less than 1 ppm, which is reasonable according to the present experiments.

4.2. Blending effects of H_2 on NH_3 oxidation

4.2.1. Effects of H_2 blending on fuel conversion

The bond dissociation energy (BDE) of H_2 is 104.15 kcal/mol [83], while that of N-H bond in the NH_3 molecule is 107.57 kcal/mol [84]. Therefore, chain-initiation reactions in H_2 oxidation can occur at lower temperatures than in NH_3 oxidation. When blending H_2 in NH_3 oxidation, the chain-initiation reactions first occurs in H_2 oxidation at lower temperatures, and the radicals produced in H_2 oxidation can simultaneously trigger the NH_3 oxidation. Therefore, the temperature window for NH_3/H_2 oxidation is lower than for pure NH_3 oxidation. To better understand the interactions between the sub-mechanisms of H_2 and NH_3 , the rate of production (ROP) and sensitivity analyses of NH_3 in the oxidation of pure NH_3 and NH_3/H_2 mixtures are conducted at the same conversion ratio of NH_3 with the present model. The chemical effects of H_2 blending can be revealed.

In all the conditions investigated, and based on model analyses, OH radical is the dominant radical to consume NH_3 . More OH radical yields can enhance the consumption of NH_3 . In pure NH_3 oxidation, as seen from Fig. 5, R15 ($NH_2 + NO = NNH + OH$) and R16 ($NH_2 + NO = N_2 + H_2O$) have the largest negative and positive sensitivity coefficients for NH_3 consumption, respectively, since they are the dominant chain-branching and chain-termination reactions, respectively. These two pathways are also shown in Fig. 6 (blue dashed lines). Through these pathways, an OH radical can be produced, which can subsequently consume NH_3 .

With increasing H_2 content, R3 ($NH_3 + H = NH_2 + H_2$) proceeds in its reverse direction—the NH_2 radical combines with H_2 to produce the NH_3 and H atom. This reaction is only sensitive in NH_3/H_2 mixtures while not sensitive in pure NH_3 oxidation at $\phi = 0.25$ (see Fig. 5). It has two effects: The blending H_2 is converted to NH_3 , compensating the decreasing content of NH_3 ; the NH_2 rad-

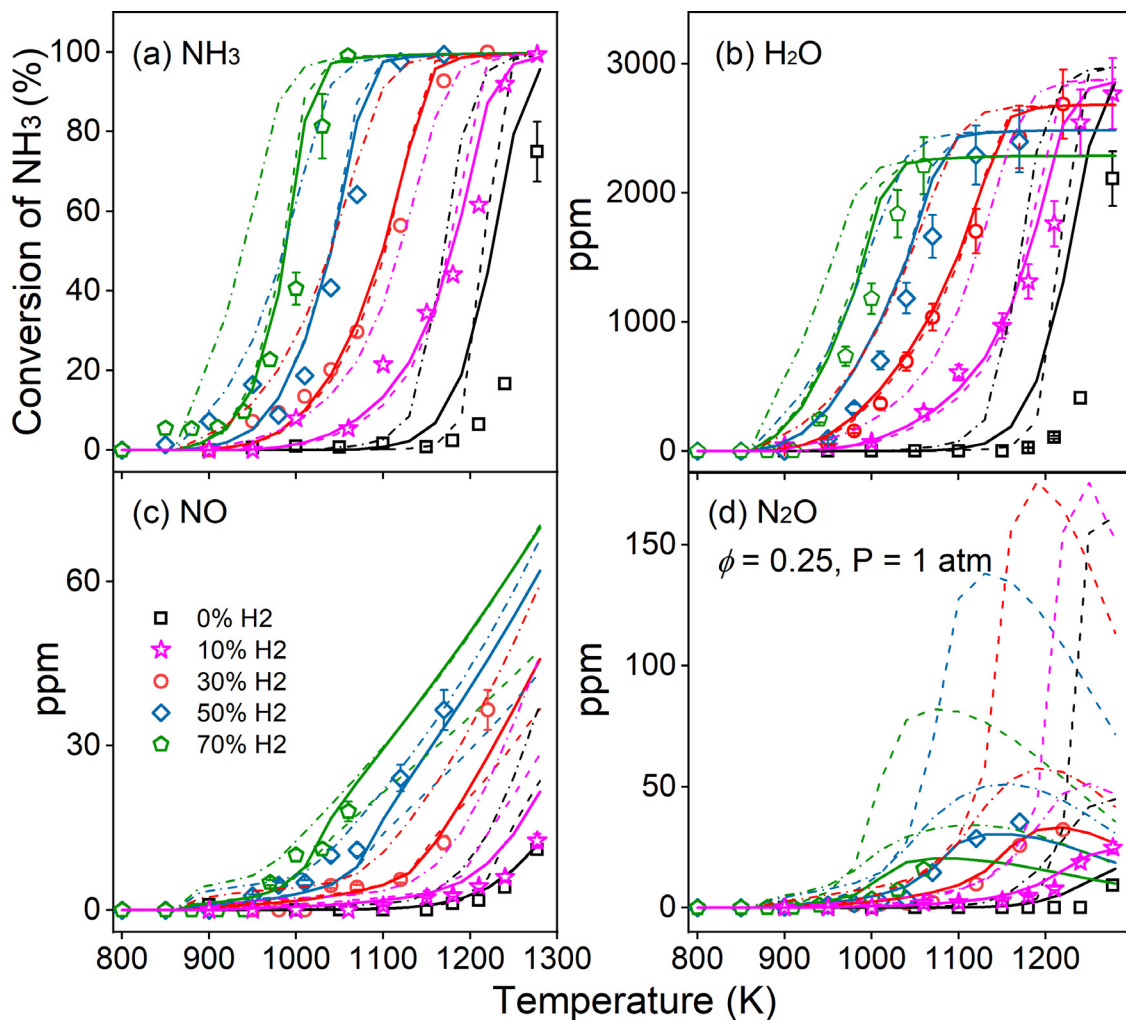


Fig. 3. Measured and simulated conversion ratio of (a) NH_3 and mole fractions of (b) H_2O , (c) NO and (d) N_2O in JSR oxidation of NH_3/H_2 mixtures with various H_2 content at $\phi = 0.25$, $\tau = 1$ s and $P = 1$ atm. Symbols are measured results from the present work. Lines are simulated results by different models. Solid, dashed and dash-dot lines are the present, Polimi 2020 [49] and Glarborg 2018 [31] models, respectively.

ical is converted to an H atom, which can combine with O_2 to proceed a new chain-branching pathway via R33. R33 is the specific chain-branching reaction in H_2 oxidation, which presents the largest negative sensitivity for NH_3 consumption in NH_3/H_2 oxidation under the present conditions, as seen from Fig. 5. As a competitive reaction pathway, R34 converts an H atom to HO_2 radical, which is responsible for chain propagation under the conditions investigated here. In summary, H_2 blending enhances the reverse reaction of R3 (i.e. R-3). The H atom produced via R-3 triggers the most important chain-branching and chain-propagation reactions in H_2 oxidation (i.e. R33 and R34), resulting in the change of dominant chain-branching reaction from R15 to R33, and in the production of more radicals, such as H, O, OH and HO_2 , as shown in Fig. 6 (red lines).



HO_2 is an important radical at intermediate temperatures. The increasing content of H_2 enhances its production via R34. It can combine with NH_2 to produce H_2NO and OH via R23 ($\text{NH}_2 + \text{HO}_2 = \text{H}_2\text{NO} + \text{OH}$), which is competitive with DeNO_x pathways of R15 and R16. With increasing H_2 content, more HO_2 production makes DeNO_x pathways less favorable than R23. H_2NO

formed via R23 can then react with the H atom to produce NH_2 and OH via R35. In this reaction sequence, NH_2 and H_2NO are served as chain carriers, converting one HO_2 and one H to two OH radicals, as seen from the solid blue lines in Fig. 6. As mentioned above, because OH radical is the dominant radical to consume NH_3 , the enhanced OH formation can promote NH_3 consumption. On the other hand, the enhanced formation of HO_2 can promote the reverse reaction of R7 ($\text{NH}_3 + \text{O}_2 = \text{NH}_2 + \text{HO}_2$), resulting in the chain-termination and the inhibition of the reaction systems.



4.2.2. Effects of H_2 blending on NO_x formation

To better compare the NO_x formation trend with increasing H_2 content, the measured and predicted mole fractions of NO, N_2O and the sum of NO and N_2O at the 99% conversion of NH_3 are plotted in Fig. 7. Note that we did not measure the speciation data at $\omega(\text{NH}_3) = 99\%$ in pure NH_3 oxidation. Only the predicted NO_x concentrations by the present model are shown in Fig. 6 for cases of pure NH_3 . As seen from Fig. 7a, under fuel-lean conditions, the measured NO, N_2O and their total concentrations first increase and then decrease with increasing H_2 content. The peak value of NO_x (including NO and N_2O in this work) seems to locate between $x(\text{H}_2) = 30\%$ and $x(\text{H}_2) = 50\%$. In contrast, more complex NO_x formation trends are observed under stoichiometric conditions, as

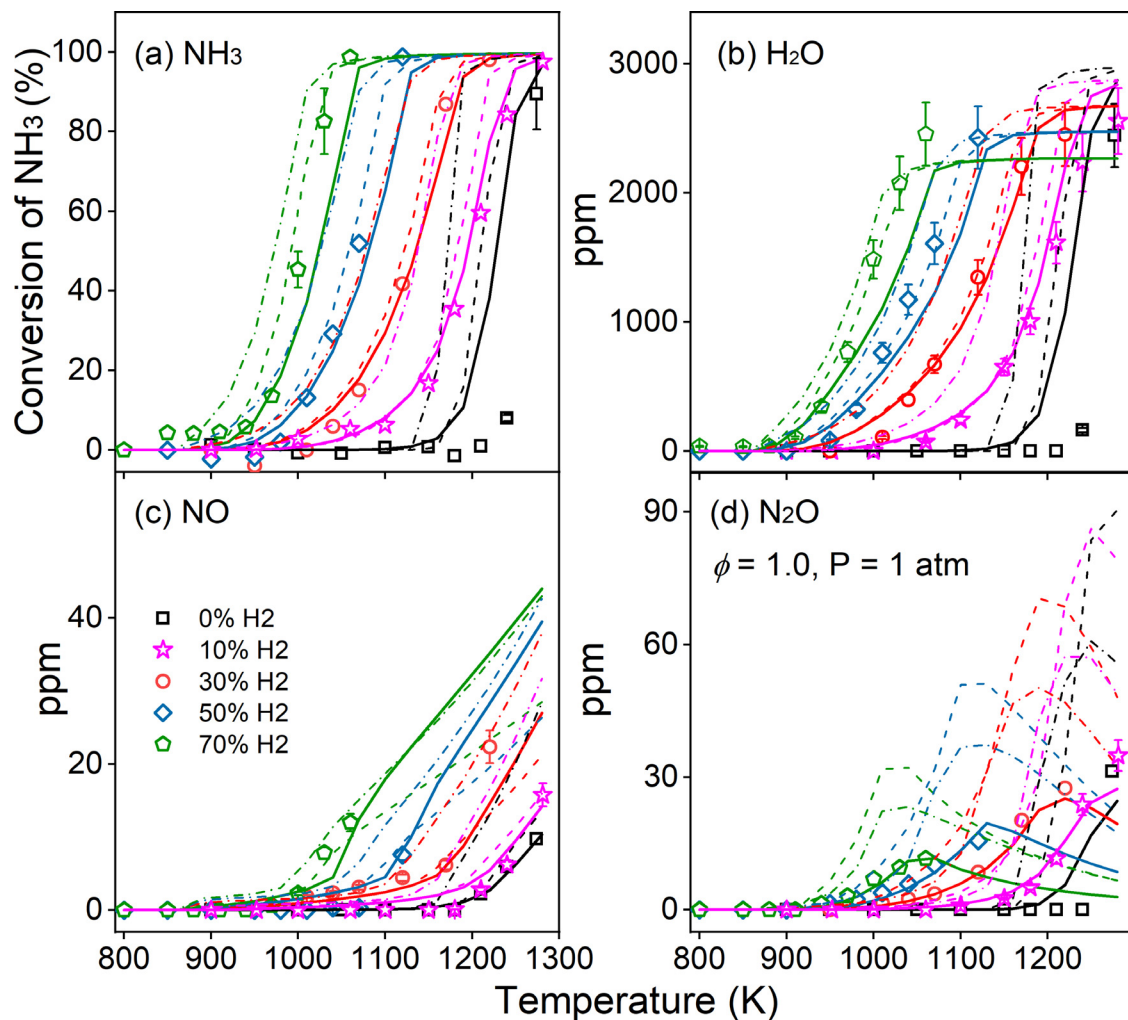


Fig. 4. Measured and simulated conversion ratio of (a) NH_3 and mole fractions of (b) H_2O , (c) NO and (d) N_2O in JSR oxidation of NH_3/H_2 mixtures with various H_2 content at $\phi = 1.0$, $\tau = 1$ s and $P = 1$ atm. Symbols are measured results from the present work. Lines are simulated results by different models. Solid, dashed and dash-dot lines are the present, Polimi 2020 [49] and Glarborg 2018 [31] models, respectively.

seen from Fig. 7b. The measured N_2O formation continues to decrease with the addition of H_2 , while the yield of NO seems to reach its minimum at $x(\text{H}_2) = 50\%$. As a result, the minimum mole fraction of NO_x (including NO and N_2O in this work) appears to locate between $x(\text{H}_2) = 50\%$ and $x(\text{H}_2) = 70\%$. In general, the present model can reasonably capture the formation trends of NO_x with H_2 blending. Under fuel-lean conditions, and compared with pure NH_3 oxidation (dashed horizontal line), blending H_2 increases NO_x formation at $\omega(\text{NH}_3) = 99\%$, as seen from Fig. 7a. However, under stoichiometric conditions, the predicted NO_x in H_2 blending oxidation can be lower than in pure NH_3 oxidation, when $x(\text{H}_2) > 50\%$. In the present experiments, heat releases are neglected and the reactor can be approximately considered as an adiabatic system. Therefore, the thermal effects introduced by H_2 addition can be neglected; only dilution and chemical effects are discussed in this work.

Under the present conditions, the thermal NO_x pathways that lead to the formation of NO are considered negligible, due to the intermediate-temperature conditions. In contrast, the fuel- NO_x pathways in NH_3 oxidation play a dominant role in NO formation. For pure H_2 oxidation under the present conditions, the NO_x formation via thermal NO_x pathways is neglected, and there are no fuel- NO_x pathways. Therefore, blending H_2 can reduce NO_x formation because of the dilution effects. However, considering only the dilution effects cannot explain the non-monotonic NO_x formation

trends observed in Fig. 7; the chemical effects of H_2 addition must play an important role.

To illustrate the chemical effects of H_2 blending, the reaction pathways of NH_3 oxidation and NO_x formation are analyzed, based on the ROP analyses. As seen from Fig. 8, consumption of NH_3 can be divided into NO_x formation related pathways (red box), (oxidative) pyrolysis pathways (yellow box) and De NO_x related pathways (green box). NH_2 , NH and N radicals can either combine with O_2 , O , OH and HO_2 to proceed via NO_x formation related pathways like R21, R23-R27, or react with NO and NO_2 to undergo De NO_x related pathways like R15-R20. Therefore, NO_x formation and De NO_x related reactions compete with each other. As mentioned above, the increasing content of H_2 boosts the production of oxygenated radicals (i.e. O , OH and HO_2). These radicals can make NO_x formation pathways more competitive than the De NO_x pathways, resulting in the enhancement of NO formation. Therefore, H_2 blending has enhancing chemical effects on NO formation. These chemical effects can explain the increasing NO formation trend with increasing H_2 content from zero % to 30%, as shown in Fig. 7. The formation of N_2O is mainly through R18 ($\text{NH}_2 + \text{NO}_2 = \text{N}_2\text{O} + \text{H}_2\text{O}$) and R19 ($\text{NH} + \text{NO} = \text{N}_2\text{O} + \text{H}$). On one hand, increasing H_2 content makes the De NO_x related pathways less favorable, suppressing the reactions of R18 and R19. On the other hand, enhanced production of NO and NO_2 caused by increasing H_2 content, promotes the reactions of R18 and R19. Therefore, H_2 blend-

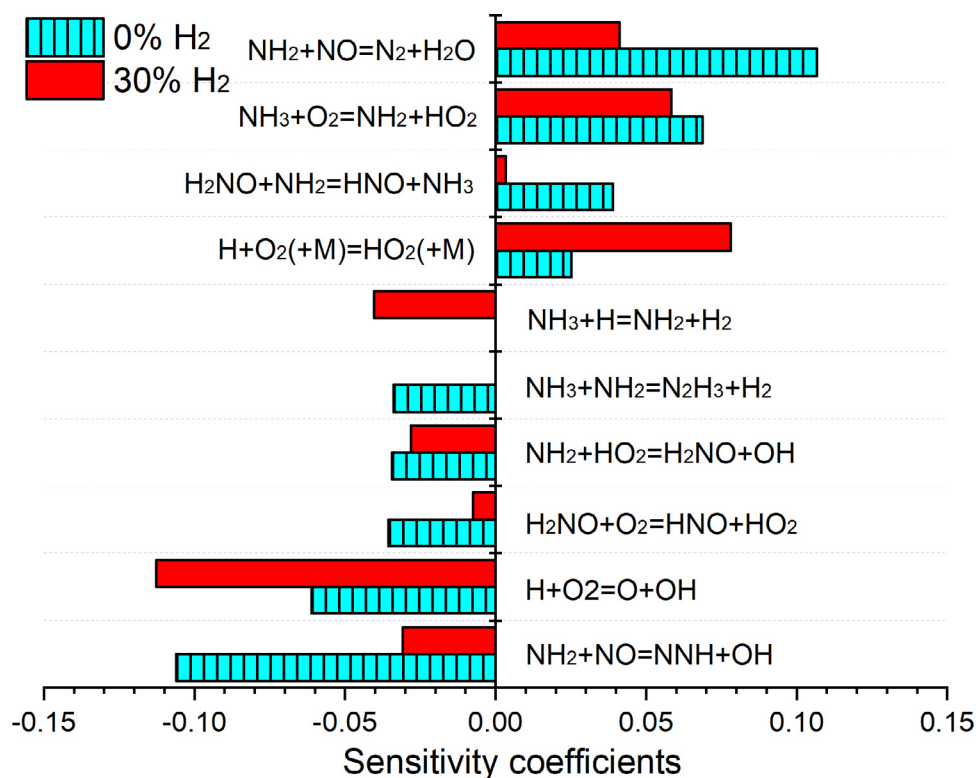


Fig. 5. Sensitivity analysis of NH_3 in the JSR oxidation of pure NH_3 (blue bars) and 70% NH_3 /30% H_2 mixture (red bars) at $\phi = 0.25$ and $\omega(\text{NH}_3) = 5\%$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

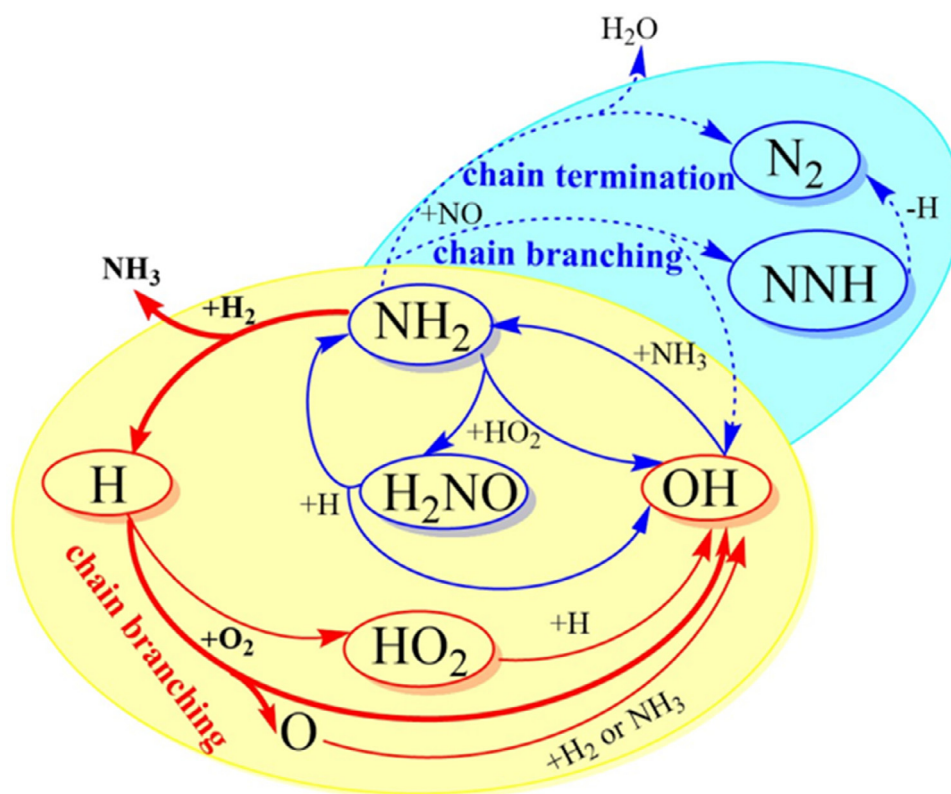


Fig. 6. H_2 blending effects on NH_3 consumption. Dashed arrows represent most important chain-branching and chain-termination pathways in pure NH_3 oxidation. Solid arrows are reaction pathways enhanced by H_2 blending. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

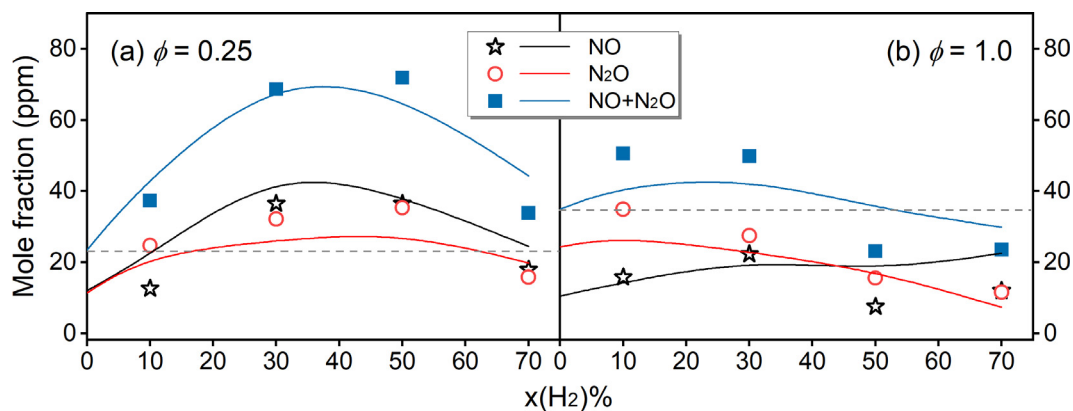


Fig. 7. Measured and predicted mole fractions of NO, N₂O and sum of NO and N₂O with increasing H₂ content at $\omega(\text{NH}_3) = 99\%$. Symbols are present experimental data. Solid lines are simulated results from present model. Dashed horizontal lines are simulated NO+N₂O mole fractions in pure NH₃ oxidation.

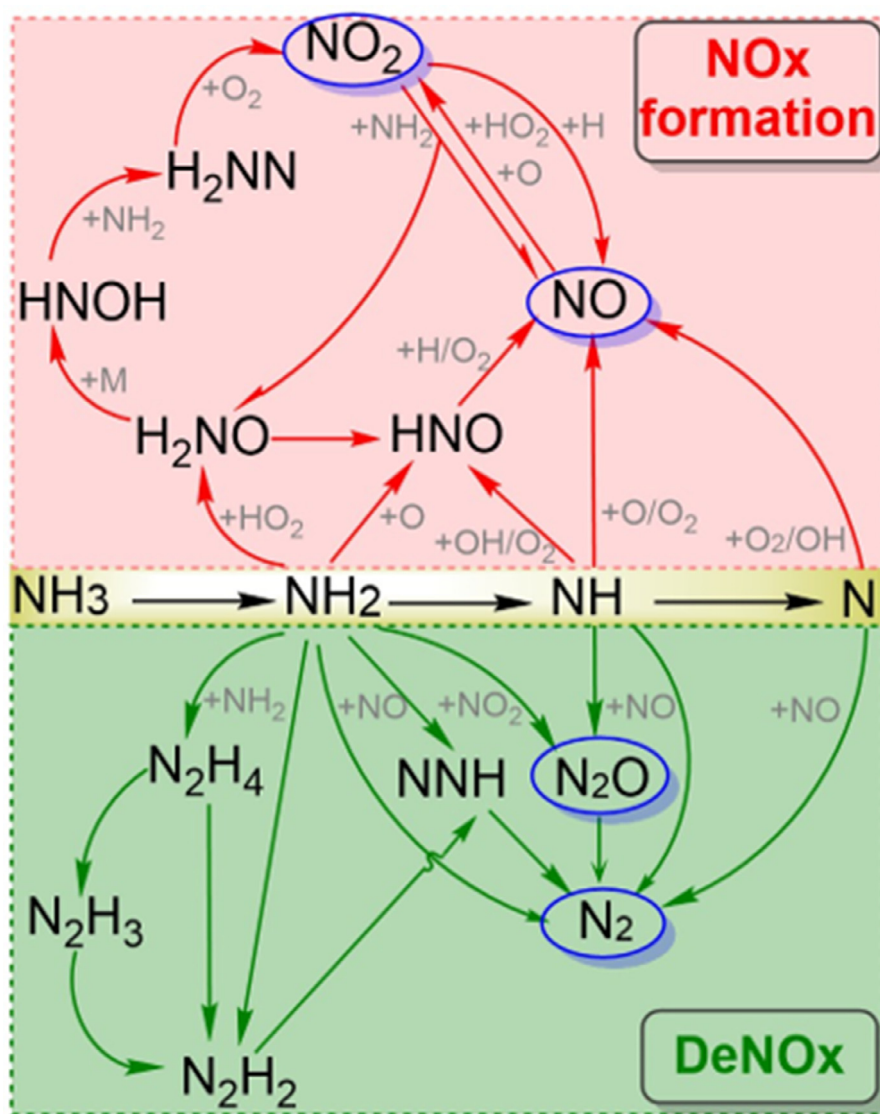


Fig. 8. NO_x formation and DeNO_x pathways in JSR oxidation of NH₃. Red and green boxes highlight reaction pathways involved in NO_x formation and DeNO_x processes, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

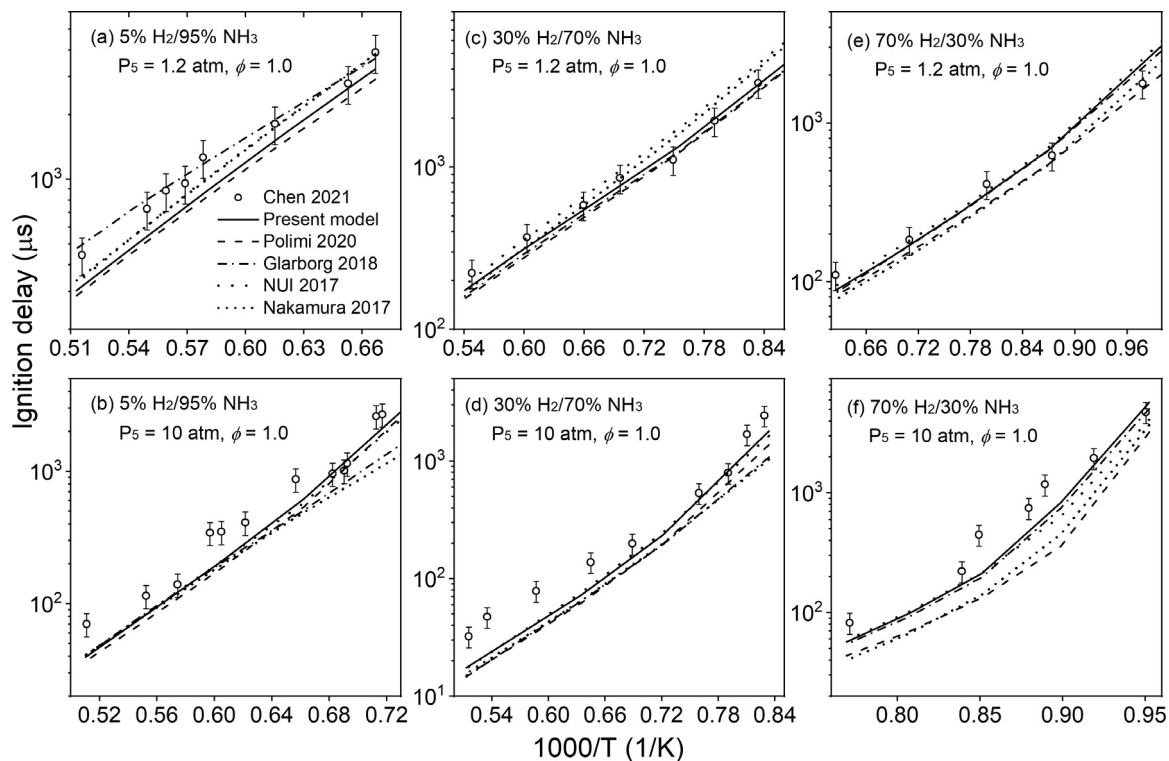


Fig. 9. Ignition delay times of NH_3/H_2 mixture measured in a shock tube at $\phi = 1.0$ and $P_5 = 1$ and 10 atm. Symbols are experimental data from Chen et al. [85]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are simulated results from the present, Polimi 2020 [49], Glarborg 2018 [31], NUI 2017 [30] and Nakamura 2017 [82] models, respectively.

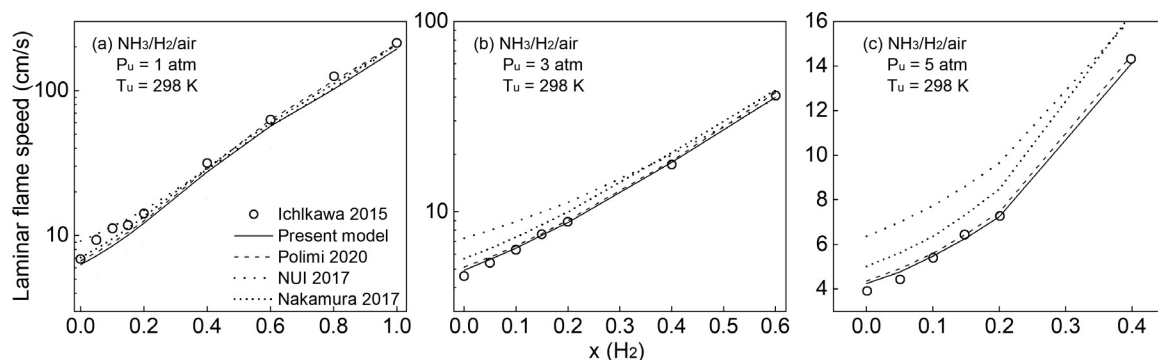


Fig. 10. Laminar flame speeds of NH_3/H_2 mixtures as a function of H_2 content at (a) $P_u = 1$ atm, (b) $P_u = 3$ atm and (c) $P_u = 5$ atm and $T_u = 298$ K and $\phi = 1.0$. Symbols are experimental data from Ichikawa et al. [17]. Solid, dashed, sparse dotted and dense dotted lines are simulated results of the present, Polimi 2020 [49], NUI 2017 [30] and Nakamura 2017 [82] models, respectively.

ing has both suppressing and enhancing chemical effects on N_2O formation.

In summary, in the present work, both dilution and chemical effects of H_2 addition contribute to NO_x formation. With increasing H_2 content, the dilution effects from the H_2 addition continue to reduce NO_x formation. The chemical effects of H_2 blending promote the formation of NO . In Fig. 7 (i.e. the case for $\omega(\text{NH}_3) = 99\%$), chemical effects seem to play a dominant role in NO formation, since the predicted mole fractions of NO in H_2 blending cases are all higher than those in pure NH_3 oxidation. Chemical effects can both suppress and enhance N_2O formation. Under fuel-lean conditions (Fig. 7a), the enhancing chemical effects seem to be more important, since the simulated mole fractions of N_2O in the cases of H_2 addition are all higher than those in pure NH_3 oxidation. However, under stoichiometric conditions (shown in Fig. 7b), the predicted N_2O yields in the cases of $x(\text{H}_2) > 10\%$ are less than in pure NH_3 oxidation, indicating that the dilution

effects, and/or the suppressing chemical effects of H_2 addition, become more important than the enhancing chemical effects.

4.3. Additional model validations

4.3.1. Ignition delay times of NH_3/H_2 mixtures

In addition to validations against the experimental targets obtained in this work, the present model is also validated against the experimental data of NH_3/H_2 mixtures in the literature. Chen et al. [21] recently measured the ignition delay times of stoichiometric NH_3/H_2 mixtures in a shock tube at both 1 and 10 atm. The H_2 content ranged from zero to 70 vol%. Their measured results are selected as validation targets at intermediate and high temperatures. Comparison between their measured and present simulated results is presented in Fig. 9. Also, the results predicted by recent NH_3 models [30,31,49,82] from different research groups are selected for comparison. As seen from Fig. 9, the present model

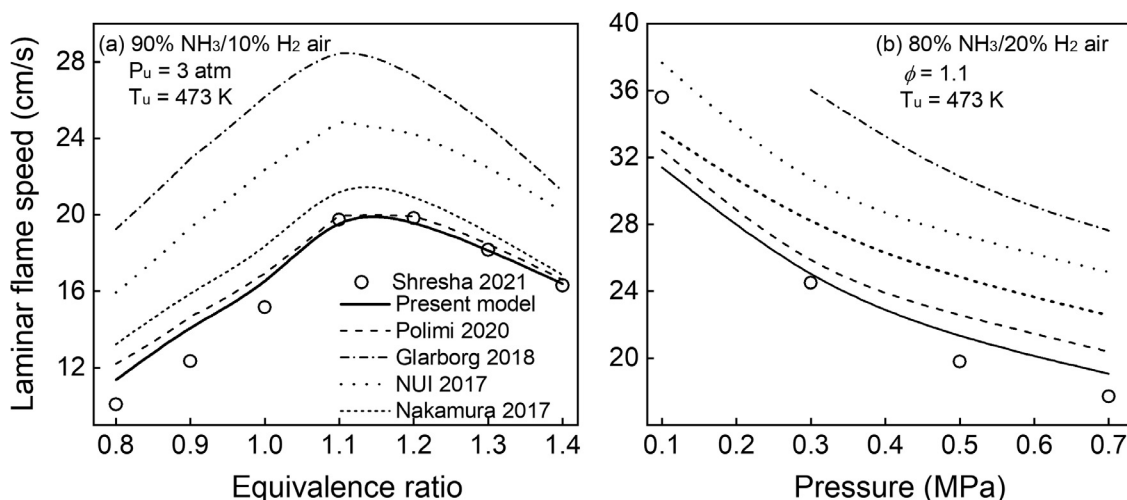


Fig. 11. Laminar flame speeds of NH_3/H_2 mixtures at (a) $T_u = 473$ K and $P_u = 3$ atm and (b) $\phi = 1.0$ and $T_u = 473$ K. Symbols are experimental data from Shresha et al. [14]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are simulated results of the present, Polimi 2020 [49], Glarborg 2018 [31], NUI 2017 [30] and Nakamura 2017 [82] models, respectively.

(solid lines) can satisfactorily capture the experimental data. In addition to the ignition delay times measured in the shock tubes, those obtained in RCMs at higher pressures and lower temperatures are also selected as validation targets for the present model. Comparison between the measured and predicted ignition delay times is presented in Fig. S37 in the *Supplementary Material*. The experimental data are taken from He et al. [24] with the H_2 content ranging from 1 to 20 vol% and pressures of 20 and 40 atm. Like previous models [30,31,49,82], the present model generally over-predicts the ignition reactivity when $x(\text{H}_2) > 5\%$. Based on the model analyses, the H_2NO related reactions play an important role in predicting ignition under these conditions, while their related kinetic parameters still have significant uncertainties. The present model will be further improved, based on future theoretical calculation progress on these reactions.

4.3.2. Laminar flame speeds of NH_3/H_2 mixtures

The present model is also validated against the laminar flame speeds of NH_3/H_2 mixtures. Figure 10 presents the measured and predicted laminar flame speeds of NH_3/H_2 mixtures as a function of H_2 content at 1, 3 and 5 atm. Experimental data are taken from Ichikawa et al. [17]. As seen from Fig. 10, the present model captures their measured results well. Compared with predictions by previous models [30,49,82], the present and Polimi 2020 models better predict laminar flame speeds obtained at 5 atm. Figure 11 compares the measured and predicted laminar flame speeds as a function of equivalence ratio and pressure at an unburned temperature of 473 K. Experimental data are from Shresha et al. [14]. As seen from Fig. 11, the present model can capture the laminar flame speeds at elevated pressures better than the previous models [30,31,49,82]. Additional validation results can be found in Figs. S38 and S39 in the *Supplementary Materials*. Compared with previous models [30,49,82], the present model also captures laminar flame speeds measured under fuel-rich conditions better (see Fig. S38 in the *Supplementary Materials*), and reasonably predicts the laminar flame speeds measured at elevated temperatures (see Fig. S39 in the *Supplementary Materials*). Flame speciation data of NH_3/H_2 mixtures, measured by Duynslaegher et al. at low pressures [25], are also selected to validate the present model. The validation results of the present model are presented in Fig. S40 in the *Supplementary Materials*. Generally, the present model reasonably predicts the mole fractions of NH_3 , H_2 , N_2 and NO within the experimental uncertainties.

5. Conclusions

In this work, JSR oxidation experiments on NH_3/H_2 mixtures were conducted at atmospheric pressure. The H_2 content in fuel mixtures ranged from zero to 70 vol% at equivalence ratios of 0.25 and 1.0. Species identification and quantification were achieved using online FTIR. Mole fractions of NH_3 , H_2O , NO and N_2O were measured in this work for model validation. The experimental results in the present work show that H_2 blending promotes the conversion of NH_3 at given temperatures and lowers the onset temperature of NH_3 oxidation. For NO_x formation, non-monotonous trends were observed by increasing the content of H_2 at the 99% conversion of NH_3 . In addition, a kinetic model for NH_3/H_2 oxidation was developed in this work and validated against the experimental data involved in H_2/NO , $\text{H}_2/\text{N}_2\text{O}$, NH_3/NO , NH_3/NO_2 , pure NH_3 and NH_3/H_2 mixtures. The present model satisfactorily predicts the present experimental data, as well as the data in the literature.

The chemical effects of H_2 blending on NH_3 oxidation were revealed, based on the model analyses. The increasing H_2 content causes R3 to proceed in its reverse direction, converting the NH_2 radical to H atom. The H atom can trigger R33 and R34 to yield O, OH and HO_2 radicals. The most important chain-branching reaction changes from R15 to R33 with increasing H_2 content. In addition, NH_2 and H_2NO radicals can serve as chain carriers, converting one HO_2 and one H radicals to two OH radicals. Enhanced OH production promotes the consumption of NH_3 . Although the dilution effects of H_2 blending reduce NO_x formation, its chemical effects can increase NO formation, due to the enhanced production of O, OH and HO_2 radicals. In contrast, the enhanced production of O, OH and HO_2 radicals makes the DeNO_x related pathways less competitive, suppressing the formation of N_2O . However, the enhanced NO formation can promote N_2O formation via R19. Therefore, H_2 blending has both suppressing and enhancing chemical effects on N_2O formation. The final N_2O formation is determined by the competition among both aspects of the chemical effects and the dilution effects.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2021.111653](#).

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